# Paper II – Urology (MCh Urology, Summer 2018)

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## II-1. Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male.

**Type:** LAQ (25 marks)

### Introduction

- GU (genitourinary) TB (tuberculosis) is caused by the *Mycobacterium tuberculosis* complex (MTBC). It is the second commonest extrapulmonary site after lymph nodes. Untreated, it destroys the kidney and contracts the bladder.

### Pathophysiology

- **Primary infection:** inhaled bacilli spread by lymph and blood, before immunity develops, and seed the kidney and epididymis.
- **Granuloma:** macrophages, epithelioid and giant cells inside a lymphocyte cuff, kept alive by IFN-γ (interferon-γ). In 90–95% it contains the infection; viable bacilli stay dormant. Reactivation follows old age, diabetes, renal failure, HIV (human immunodeficiency virus) or steroids, sometimes decades later. Then **caseation** releases bacilli into urine.

| Route | Lesion |
|---|---|
| **Haematogenous** (principal) | Parenchymal disease: kidney, epididymis, prostate |
| **Descending urinary** | Luminal disease: ureteric stricture, cystitis, urethra |
| **Ascending** (intravesical BCG [bacille Calmette–Guérin]) | Cystitis, prostatitis, epididymo-orchitis |

| Site | Frequency | Pathology |
|---|---|---|
| Kidney | 80% | Periglomerular granulomas, then caseating cavities, papillary necrosis, calcification; **autonephrectomy** ("putty kidney") |
| Ureter | — | Stricture, commonest at the **VUJ** (vesicoureteric junction); "beaded corkscrew" or "pipestem" ureter |
| Bladder | — | Starts at the ureteric orifices ("golf-hole" orifice); ulcers; after about a year of fibrosis it contracts; **thimble bladder <20 mL** |
| Epididymis | Commonest genital site | Starts in globus minor; beaded vas, posterior scrotal sinus |

| Site | Frequency | Pathology |
|---|---|---|
| Prostate, seminal vesicles | 22–49% | Calcification, abscess; infertility |

- **Key concept:** drugs cure by fibrosis, and fibrosis obstructs. Strictures and bladder contraction can **worsen during treatment**.

## Diagnosis (brief)

- **Sterile pyuria** with or without haematuria (>90%).
- **Urine AFB (acid-fast bacilli) culture** (gold standard): 3–5 first-void morning samples on consecutive days. LJ (Löwenstein–Jensen) solid medium takes 4–6 weeks; use a liquid system as well. Culture gives drug susceptibility.
- **GeneXpert MTB/RIF:** rapid DNA test that also detects rifampicin resistance (marker of MDR [multidrug-resistant] TB).
- **Imaging:** chest radiograph in all; CT (computed tomography) urography: calcification, phantom calyx, Kerr's kink, ureteric strictures, small bladder. Biopsy if culture is negative.

## Medical treatment (first step in all)

- **2HRZE/4HR:** isoniazid, rifampicin, pyrazinamide, ethambutol for 2 months, then isoniazid and rifampicin for 4 months, with pyridoxine. Some extend to 9–12 months in complicated disease.
- Repeat ultrasound during treatment for worsening obstruction; stent or PCN (percutaneous nephrostomy) early. Steroids: anecdotal only.
- **Surgery after at least 4–6 weeks of ATT (antituberculous therapy)**, when inflammation and bacillary load have settled.

## Assessment of a thimble bladder (30-year-old man)

| Assess | How | Why |
|---|---|---|
| Disease activity | Repeat urine AFB culture; bladder biopsy | Do not reconstruct with active TB |
| Capacity and compliance | Cystogram, cystoscopy under anaesthesia, urodynamics | Size and type of reconstruction |
| Upper tracts | Ultrasound, CT urography, MCU (micturating cystourethrogram) | VUJ stricture or reflux needs reimplantation |
| Renal function | Creatinine, GFR (glomerular filtration rate), split function | Bowel reservoir needs creatinine <2 mg/dL, GFR >35–40 mL/min |
| Urethra, bladder neck, sphincter | Urethrogram, endoscopy | Orthotopic reservoir needs a patent, continent outlet |
| Patient | Dexterity, motivation, fertility wishes | May need CIC (clean intermittent catheterisation) |

## Surgical options

| Bladder state | Operation |
|---|---|
| Half the bladder diseased, capacity <100 mL | **Augmentation ileocystoplasty** ("clam" or onlay of detubularised ileum) |
| Severely contracted | Augmentation with **ileocaecum** or sigmoid |
| **Thimble bladder (<20 mL)** | **Substitution cystoplasty:** supratrigonal excision of the diseased bladder, keeping trigone and bladder neck, and an orthotopic detubularised ileal or ileocaecal reservoir |

- **Principles:** detubularise and fold the bowel into a spherical, low-pressure reservoir (Laplace's law; interrupts peristalsis). Anastomose widely to the trigone.
- **Ureters:** if the VUJ is strictured or refluxing, excise to healthy ureter and **reimplant into the reservoir**: an antireflux tunnel under a colonic taenia, or an isoperistaltic afferent ileal limb. A Boari flap or psoas hitch is not possible in a contracted bladder.
- **Avoid orthotopic reconstruction** with: poor renal function; strictured or incompetent urethra; inability to perform CIC; active TB; short bowel or inflammatory bowel disease. Then use a continent catheterisable pouch (Mitrofanoff) or an **ileal conduit**.
- **Also know:** laparoscopic and robotic augmentation; transurethral incision of bladder neck contracture.

## Complications

- **Mucus** and catheter blockage; **stones**; bacteriuria and UTI (urinary tract infection); **hyperchloraemic metabolic acidosis**; vitamin B12 deficiency if ileum is lost; **perforation** (needs regular emptying); incomplete emptying needing CIC; nocturnal incontinence; late tumour in the reservoir.

## Follow-up

- Urine AFB culture, creatinine, bicarbonate, B12 and ultrasound. GU TB relapses in 6–22% (mean about 5 years), so follow up for up to 10 years.

**Diagram to draw:** supratrigonal excision of a thimble bladder with an ileocaecal substitute and ureteric reimplantation.

> **Key points to remember:**
> - Kidney 80%; bladder disease descends from the kidney; thimble bladder <20 mL.
> - Urine AFB culture: 3–5 early-morning samples; GeneXpert detects rifampicin resistance.
> - 2HRZE/4HR; operate after 4–6 weeks of ATT.
> - Augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.

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# II-2. Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis.

**Type:** LAQ (25 marks)

## Lymphatic drainage of the testis

- **Why retroperitoneal:** the testis develops beside the kidney (L1–L2) and its lymph returns to where it began.
- **Route:** lymphatics of the testis and the caput and corpus of the epididymis run up the cord with the testicular vessels, through the internal ring and along the gonadal vessels to the lumbar nodes near the renal hilum.
- From there lymph passes to the **cisterna chyli** (in front of L1–L2), the **thoracic duct** and the left innominate vein. This explains **left supraclavicular** node metastases.

**Retroperitoneal nodal regions**

| Region | Extent |
|---|---|
| Left para-aortic | Aortic midline to the left ureter |
| Interaortocaval | Between IVC (inferior vena cava) and aorta |
| Right paracaval | IVC midline to the right ureter |

**Primary landing zones**

| | Right testis | Left testis |
|---|---|---|
| Primary | **Interaortocaval** | **Left para-aortic** (and left hilar) |
| Secondary | Right paracaval, precaval | Interaortocaval, pre-aortic |
| Crossover | Small but real to the left side | Rare to the right, unless bulky |

- **Crossover is right to left** only, because lymph flows from right to left.
- **Cauda of epididymis** drains with the vas to the **external iliac** nodes.
- **Pelvic and inguinal nodes** are involved only if normal drainage is altered: previous scrotal or inguinal surgery (orchidopexy, hernia repair), scrotal violation, or tumour invading the tunica vaginalis or scrotum.

## Lymphatic drainage of the scrotum

- Scrotal skin drains to the **ipsilateral superficial inguinal nodes**. Lymphatics, like vessels and nerves, do not cross the median raphe.

## Clinical relevance

- **Trans-scrotal orchidectomy or biopsy is contraindicated.** It leaves cord behind and opens scrotal lymphatic routes, causing local and inguinal or pelvic recurrence.
- Explains RPLND (retroperitoneal lymph node dissection) templates and the para-aortic or "dog-leg" radiotherapy fields.

## Stage I seminoma: definition and staging

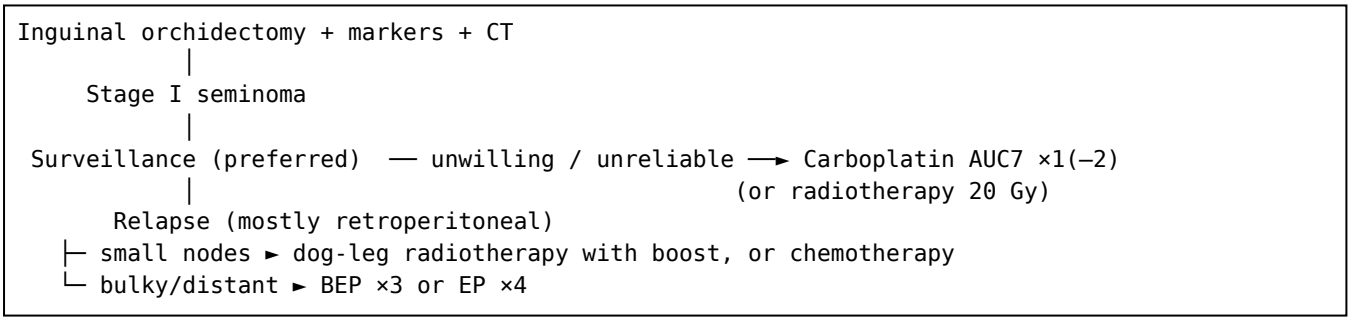
- Disease confined to the testis: any pT, N0, M0 on imaging. Stage IA = pT1; IB = pT2–4; IS = markers still raised after orchidectomy.
- **Work-up:** AFP (alpha-fetoprotein), β-hCG (human chorionic gonadotropin) and LDH (lactate dehydrogenase) before and after orchidectomy. CT abdomen and pelvis; chest CT or radiograph. Baseline testosterone; **offer sperm banking**. A raised AFP means non-seminoma.

- **Radical inguinal orchidectomy:** early cord control at the internal ring. It cures 80–85% of stage I seminoma.
- **Risk factors for relapse: tumour >4 cm** and **rete testis invasion**. Five-year relapse is 12%, 16% and 32% with 0, 1 and 2 factors. LVI (lymphovascular invasion) predicts relapse in non-seminoma, not seminoma.

## Options after orchidectomy (cancer-specific survival about 100% with each)

| | Surveillance | Carboplatin | Radiotherapy |
|---|---|---|---|
| Method | Clinical review, markers, CT | **One cycle, AUC (area under the curve) 7** | **20 Gy** in 10 fractions to the para-aortic strip |
| Relapse | 14–20% | About 5% | 3–5% |
| Relapse site | Retroperitoneum (84–100%) | Retroperitoneum | Pelvis, chest, left supraclavicular |
| Drawbacks | Late relapse, many scans, compliance | Underdosing; late effects unknown | Second cancers, GI (gastrointestinal) toxicity, oligospermia |
| Status | **Preferred** | If unwilling or unsuitable for surveillance; risk factors | If unwilling or unsuitable for surveillance |

- **Carboplatin (Calvert):** 7 × (GFR [glomerular filtration rate] + 25) mg, using isotope GFR.
- **Trials: MRC TE18** (Medical Research Council): 20 Gy equals 30 Gy, with less toxicity. **MRC TE19/EORTC (European Organisation for Research and Treatment of Cancer) 30982:** one cycle of carboplatin equals radiotherapy (5-year relapse-free 94.7% vs 96%). MRC TE10: para-aortic field equals dog-leg, with 2% pelvic relapse.
- **Risk-adapted (EAU [European Association of Urology]):** surveillance for low risk; with risk factors discuss surveillance or carboplatin (treating all would overtreat two-thirds).




- BEP: bleomycin, etoposide, cisplatin; EP: etoposide, cisplatin.

## Relapse and IGCCCG

- Relapses are almost always cured; median 14 months, but 10–20% occur after 4 years.
- **IGCCCG** (International Germ Cell Cancer Collaborative Group): seminoma is good risk unless there are non-pulmonary visceral metastases (intermediate). **There is no poor-risk seminoma**, and markers do not change the risk group.

## Follow-up (surveillance)

- Examination and markers every 2–4 months (years 1–3), 6-monthly (years 4–7), then yearly; CT every 4–6 months (years 1–3), then 6-monthly to year 7, then yearly. CT finds >85% of relapses.

## Recent advances

- **miR-371a-3p** serum microRNA detects relapse better than standard markers.

- MRI (magnetic resonance imaging) instead of CT for surveillance (TRISST trial) to reduce radiation.

**Diagram to draw:** retroperitoneal nodes with right (interaortocaval) and left (para-aortic) landing zones and the right-to-left crossover arrow.

**Key points to remember:**

- Right testis to interaortocaval; left to para-aortic; crossover only right to left.
- Scrotum to ipsilateral superficial inguinal nodes; never operate trans-scrotally.
- Risk factors: >4 cm and rete testis invasion (not LVI).
- Surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).

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# II-3a. Oligospermia.

**Type:** SAQ (10 marks)

## Definition

- **Oligozoospermia:** sperm concentration below the lower reference limit. **WHO (World Health Organization) 2021 (6th edition): 16 million/mL** (total count 39 million per ejaculate). Earlier thresholds were 20 million (1999) and 15 million (2010).
- These are 5th centile values of fertile men, not fertility cut-offs. A single value must be read with testis size and FSH (follicle-stimulating hormone).
- **Severity:** mild to moderate 5–16 million/mL; **severe <5 million/mL**; cryptozoospermia (sperm only in a centrifuged pellet).

| Other WHO 2021 lower limits | Value |
|---|---|
| Volume | 1.4 mL |
| Progressive motility | 30% |
| Normal forms (strict) | 4% |
| Vitality | 54% |

## Causes

- **Pre-testicular:** hypogonadotropic hypogonadism, **exogenous testosterone or anabolic steroids**, hyperprolactinaemia, obesity (raised oestradiol).
- **Testicular: varicocele** (commonest correctable; heat and oxidative stress), cryptorchidism, mumps orchitis, torsion, chemotherapy or radiotherapy, Klinefelter mosaic, **AZFc (azoospermia factor c) deletion**; **idiopathic** (largest group).
- **Post-testicular:** partial ejaculatory duct obstruction (low volume), infection, antisperm antibodies.
- Lifestyle: smoking, alcohol, heat, drugs.

## Evaluation

- **Two semen analyses** after 2–7 days of abstinence.
- History and examination: testis volume, varicocele, vasa, secondary sex characters.
- Morning **testosterone, LH (luteinising hormone), FSH**; oestradiol and prolactin if testosterone is low.
- **<5 million/mL: karyotype and Y-microdeletion** testing (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine] guideline).
- Selected: scrotal Doppler; TRUS (transrectal ultrasonography) if volume is low; semen culture; DNA fragmentation.

## Management

- **General:** stop testosterone and gonadotoxins; weight loss; stop smoking.
- **Varicocelectomy** (microsurgical subinguinal) for a **palpable** varicocele with abnormal semen. It improves semen and pregnancy rates. Do not treat a subclinical one.
- **Medical:** hypogonadotropic: hCG (human chorionic gonadotropin) with or without FSH. Idiopathic with low testosterone: off-label clomiphene 25 mg daily, or anastrozole 1 mg daily if the testosterone to oestradiol ratio is low.
- **ART (assisted reproductive technology):** mild: IUI (intrauterine insemination); severe or failed IUI: **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Bank sperm in AZFc carriers because counts may fall.
- Spermatogenesis takes about 64 days; judge response after 3–6 months.

**Diagram to draw:** hypothalamic–pituitary–testis axis with sites of treatment.

**Key points to remember:**

- WHO 2021: 16 million/mL; severe <5 million/mL.
- Karyotype and Y-microdeletion if <5 million/mL.
- Repair palpable varicoceles only; never give testosterone.

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# II-3b. DMSA Renal Scan.

**Type:** SAQ (10 marks)

## Principle

- **$^{99m}$Tc-DMSA** (technetium-99m dimercaptosuccinic acid) is filtered and then taken up by **proximal tubular cells** (megalin and cubilin receptors). It is retained in the cortex and excreted very slowly.
- So it gives a **static map of functioning cortex**, not drainage.

## Technique

- Intravenous dose: **50 µCi/kg in children, 5 mCi (185 MBq) in adults**.
- Images at **2–4 hours** (later in renal failure): posterior and oblique views, **pinhole collimator** or **SPECT** (single-photon emission computed tomography), which is slightly more sensitive.
- Radiation about 1 mSv.

## What it measures

- **DRF (differential renal function):** from geometric-mean counts; normal 45–55%. Better than MAG3 (mercaptoacetyltriglycine) for ectopic, horseshoe or duplex kidneys and very poor function.
- **Cortical defects:** photopenic areas.

## Indications

- **Acute pyelonephritis** (gold standard), especially in infants with febrile UTI (urinary tract infection).
- **Renal scarring** in VUR (vesicoureteric reflux) and recurrent UTI; top-down approach (DMSA first, VCUG [voiding cystourethrogram] only if abnormal).
- Function of a duplex upper moiety, ectopic or horseshoe kidney; MCDK (multicystic dysplastic kidney: no uptake).
- Reconstruct or remove a poorly functioning kidney.

## Acute pyelonephritis versus scar

| | Acute pyelonephritis | Scar |
|---|---|---|
| Timing | Within 1 week of fever (sensitivity falls by 1 month) | **≥6 months** after infection |
| Defect | Ill-defined, may be multiple | Sharp, with cortical thinning |
| Kidney size | Normal or swollen | Small or contracted |
| Repeat scan | Resolves (about 85%) | Persists |

### Goldraich scar grading

| Type | Finding |
|---|---|
| 1 | No more than 2 scarred areas |
| 2 | More than 2 scars, normal tissue between |
| 3 | Generalised damage, small kidney |
| 4 | End-stage, shrunken kidney, little or no uptake |

## Limitations

- Cannot tell **congenital reflux dysplasia** (boys, grade IV–V VUR, no UTI) from acquired scar.
- Gross hydronephrosis gives false defects; no information on drainage; poor uptake in renal failure and proximal tubular disorders.
- **Also know:** MAG3 is best for obstruction; NICE (National Institute for Health and Care Excellence): late DMSA 4–6 months after atypical or recurrent UTI.

**Diagram to draw:** posterior DMSA image: normal kidney and a scarred kidney with polar defects.

> **Key points to remember:**
> - DMSA binds proximal tubules: a static cortical scan at 2–4 h.
> - Gold standard for acute pyelonephritis and scars.
> - Scar only if the defect persists ≥6 months; normal DRF 45–55%.

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## II-3c. Dutasteride.

**Type:** SAQ (10 marks)

### Pharmacology

- A **dual 5ARI (5α-reductase inhibitor)**: it blocks **type 1** (liver, skin) and **type 2** (prostate stroma, genital tissue) isoenzymes. Finasteride blocks type 2 only.
- **Mechanism:** blocks conversion of testosterone to DHT (dihydrotestosterone), the main prostatic androgen; serum DHT falls about **90%**. Epithelial atrophy shrinks the gland and relieves the **static** component of obstruction.
- **Dose: 0.5 mg once daily**; hepatic metabolism; **half-life about 5 weeks**.

### Effects

- **Prostate volume falls about 25%** (maximum by 6 months).
- Symptoms and Qmax (maximum flow rate) improve modestly and slowly (**at least 6 months**).
- **Changes natural history:** pivotal trials: AUR (acute urinary retention) down 57%, BPH (benign prostatic hyperplasia) surgery down 48%.
- **PSA (prostate-specific antigen) halves** by 6–12 months: **double the PSA** to interpret it. A rise from the nadir on treatment needs evaluation.

### Indications

- Moderate-to-severe LUTS (lower urinary tract symptoms) with an enlarged prostate: **>40 mL (EAU [European Association of Urology])** or PSA about >1.5 ng/mL; AUA (American Urological Association): >30 mL.
- **Combination with tamsulosin** for men at risk of progression (fixed-dose capsule available).
- Off-label: BPH haematuria (lowers prostatic microvessel density); before TURP (transurethral resection of the prostate) to reduce bleeding (inconsistent evidence).

### Landmark trials

| Trial | Result |
|---|---|
| **CombAT** (Combination of Avodart and Tamsulosin, 4 years; PV [prostate volume] ≥30 mL, PSA ≥1.5) | Combination best for symptoms and progression; dutasteride (and combination) beat tamsulosin for AUR and surgery |
| **REDUCE** (Reduction by Dutasteride of Prostate Cancer Events, 4 years) | Prostate cancer down about 23%, but slightly more Gleason 8–10; **not licensed for prevention** |

### Side effects

- **Sexual:** ED (erectile dysfunction), low libido, ejaculatory disorders; lessen with time.
- Gynaecomastia and breast tenderness; depression (FDA [Food and Drug Administration] warning).
- **No IFIS** (intraoperative floppy iris syndrome) and no hypotension, unlike α-blockers.

### Precautions

- **Teratogenic:** pregnant women must not handle leaking capsules. **No blood donation for 6 months** after the last dose.

**Diagram to draw:** testosterone to DHT pathway with 5α-reductase types 1 and 2 blocked by dutasteride.

**Key points to remember:**

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

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# II-3d. Penile Prosthesis.

**Type:** SAQ (10 marks)

## Indications

- ED (erectile dysfunction) not responding to, contraindicated for, or unacceptable with oral PDE5Is (phosphodiesterase type 5 inhibitors), injections or vacuum devices.
- ED with Peyronie's disease, after ischaemic priapism (corporal fibrosis), after prostatectomy, pelvic radiotherapy or diabetes.
- **Contraindications:** active infection (UTI [urinary tract infection], skin), uncontrolled diabetes, unrealistic expectations, reversible ED.

## Types

| Type | Example | Features |
|---|---|---|
| **Malleable (semirigid) rods** | Tactra (nitinol), Genesis (silver) | Simple, cheap, almost never fail; always firm; erode more in insensate men |
| 2-piece inflatable | Ambicor (discontinued) | No reservoir; poor flaccidity |
| **3-piece inflatable** | AMS 700 (CX, LGX), Titan | Cylinders, scrotal pump, reservoir; **best rigidity and flaccidity**; most popular |

## Preoperative preparation and consent

- Treat skin and urine infection; blood glucose **<200 mg/dL** on the day; stop anticoagulants.
- **Consent:** erection may be shorter; sensation and ejaculation may reduce; mechanical failure and infection may need reoperation.

## Surgical steps (3-piece)

- **Approach:** penoscrotal (most used; pump secured) or infrapubic (direct reservoir view; risk to dorsal nerve).
- Catheter; stay sutures; corporotomy; dilate proximally to the ischial tuberosity and distally (Hegar or Brooks dilators), instruments pointing dorsolaterally.
- **Goalpost test** for crossover; measure length; cylinders equal to or 1 cm shorter than measured length.
- Reservoir in the space of Retzius, or **ectopic submuscular** after pelvic surgery. Pump in a dependent dartos pouch.
- Activation at 4–6 weeks.

## Complications

- **Infection (1–3%)**: biofilm on the device. Prevent with intravenous vancomycin or cephalosporin plus gentamicin (up to 24 h), chlorhexidine–alcohol preparation, **antibiotic-coated devices** (halve infection) and **no-touch technique**. Treat by removing **all** parts; **Mulcahy salvage** (antiseptic washout and immediate new device) succeeds in 84%.
- **Mechanical failure** (commonest): about 15% at 5 years, 30% at 10.
- Intraoperative: crural perforation, crossover, urethral injury.
- Others: erosion, extrusion, reservoir erosion into bladder or bowel, autoinflation, penile shortening, **glans necrosis**.

## Outcomes

- Highest satisfaction of any ED treatment: **80–90%** for patient and partner.

- **Also know:** modelling for Peyronie's disease; early implant after ischaemic priapism preserves length.

**Diagram to draw:** 3-piece inflatable: two cylinders, scrotal pump, reservoir in the space of Retzius.

**Key points to remember:**

- 3-piece inflatable gives the best rigidity and flaccidity.
- Infection 1–3%; coated devices and no-touch technique; remove all parts.
- Satisfaction 80–90%; repairs 15% at 5 years.

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## II-3e. CAPD (Continuous Ambulatory Peritoneal Dialysis).

**Type:** SAQ (10 marks)

### Principle

- The **peritoneum** (about 1–2 $m^2$) is the membrane; dialysate dwells in the abdomen.
- **Diffusion:** urea, creatinine and potassium move into dialysate down a concentration gradient.
- **Ultrafiltration:** an osmotic agent draws water: **glucose** (1.5%, 2.5%, 4.25%) or **icodextrin** (for the long dwell).
- **CAPD:** about **4 exchanges of 2 L daily**, dwell 4–6 hours, overnight long dwell. **APD** (automated peritoneal dialysis) uses a night cycler.

### Catheter

- **Tenckhoff catheter:** silicone with **two Dacron cuffs** (deep cuff in rectus muscle; superficial cuff 2 cm from the exit), exit site pointing downward, tip in the pelvis.
- **Insertion:** open; **laparoscopic** (rectus sheath tunnelling, omentopexy, adhesiolysis); percutaneous Seldinger. Wait about 2 weeks before full volumes to prevent leaks.

### Adequacy

- Weekly **Kt/V urea ≥1.7** (peritoneal plus residual renal urea clearance × time ÷ urea distribution volume), with clinical well-being.
- **PET** (peritoneal equilibration test): high transporters do better on short dwells (APD).

### Advantages versus haemodialysis

- Home-based, no vascular access, **haemodynamically gentle**, **preserves residual renal function**. Preferred in children, poor vascular access and unstable heart disease.

### Contraindications

- **Absolute:** peritoneal adhesions or fibrosis, loss of membrane function, uncorrectable hernia, no one able to do exchanges.
- **Relative:** recent abdominal surgery, stoma, abdominal wall infection, massive polycystic kidneys, morbid obesity, severe malnutrition.

### Complications

- **Peritonitis:** diagnosis needs 2 of: (1) abdominal pain or **cloudy effluent**; (2) effluent **WCC (white cell count) >100/µL with >50% neutrophils**; (3) positive culture. Commonest: coagulase-negative *Staphylococcus* (touch contamination). Give **intraperitoneal** antibiotics covering Gram-positives (vancomycin or cefazolin) and Gram-negatives (ceftazidime or aminoglycoside). Remove the catheter for fungal or refractory peritonitis (no response in 5 days).
- **Exit-site and tunnel infection:** prevent with topical mupirocin or gentamicin.
- **Mechanical:** outflow failure (constipation, omental wrap, migration), pericatheter **leak**, **hernias** (raised abdominal pressure), hydrothorax, scrotal oedema (patent processus vaginalis).
- **Metabolic:** hyperglycaemia, weight gain, hypokalaemia, protein loss.
- **Long term:** ultrafiltration failure; **EPS (encapsulating peritoneal sclerosis)**: rare, presents as bowel obstruction.
- **Also know:** PD (peritoneal dialysis) catheter is removed after a successful transplant.

**Diagram to draw:** Tenckhoff catheter: deep and superficial cuffs, tunnel, downward exit site, tip in the pelvis.

**Key points to remember:**

- Peritoneum is the membrane; glucose or icodextrin gives ultrafiltration.
- Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.
- Peritonitis: cloudy fluid, WCC >100/µL, >50% neutrophils; give intraperitoneal antibiotics.

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# II-3f. Uroflowmetry.

**Type:** SAQ (10 marks)

## Definition

- A non-invasive urodynamic test that records the **volume of urine voided per unit time** (mL/s). It reflects detrusor contraction and outlet resistance together.

## Method

- Patient comes with a **comfortably full bladder** (normal desire) and voids in private, in the usual position.
- **Flowmeters:** weight transducer (load cell), rotating disc, capacitance dipstick.
- **Voided volume ≥150 mL** (Qmax is unreliable below 125–150 mL). Repeat 2–3 times; ask whether the void was typical.
- Measure **PVR** (postvoid residual) by ultrasound afterwards.

## Parameters

| Parameter | Meaning |
|---|---|
| **Qmax** (maximum flow rate) | Most useful value; normal men >15 mL/s, women >20 mL/s |
| Qave (average flow rate) | Volume ÷ flow time; less useful |
| Voided volume (VV) | Must be ≥150 mL |
| Flow time / voiding time | Voiding time includes interruptions |
| Time to Qmax | Normally in the first third of the void |
| Voiding efficiency | VV ÷ (VV + PVR) × 100 |

## Curve patterns

| Pattern | Suggests |
|---|---|
| **Bell-shaped** | Normal |
| **Plateau (flattened, box)** | Fixed obstruction: **urethral stricture** |
| Prolonged, low, skewed | BPO (benign prostatic obstruction) or detrusor underactivity |
| **Intermittent / straining** | Detrusor underactivity, abdominal straining or obstruction |
| Staccato | Dysfunctional voiding (sphincter overactivity) |
| Tower (high, short) | Overactive bladder |

## Interpretation

- **Qmax <10 mL/s:** obstruction likely (high specificity). **Qmax <15 mL/s:** more sensitive, less specific.
- **Nomograms** correct for volume: **Siroky** (men), **Liverpool** (men and women, centiles). Corrected Qmax = Qmax ÷ √VV.
- **Artefacts:** "wag" (moving stream), knocking the flowmeter, straining spikes. Read Qmax from the smoothed curve, not single spikes.

## Limitations

- **Cannot separate BOO (bladder outlet obstruction) from detrusor underactivity**; only a pressure–flow study can.
- Varies with volume, position, anxiety and between days.

## Uses

- BPH (benign prostatic hyperplasia) evaluation before treatment and follow-up after medical or surgical treatment.
- Follow-up after **urethroplasty** or optical urethrotomy.
- Children: dysfunctional voiding, after posterior urethral valve ablation.
- Women: voiding dysfunction before stress incontinence surgery.
- **EAU** (European Association of Urology): before surgery, pressure–flow study if Qmax >10 mL/s.

**Diagram to draw:** flow curves: normal bell, plateau (stricture), prolonged low (BPO), intermittent.

> **Key points to remember:**
> - Voided volume ≥150 mL; Qmax is the key value.
> - Plateau = stricture; intermittent = underactivity or straining.
> - Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.

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# Rapid Revision Sheet – Paper II

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**II-1. Describe pathophysiology of genitourinary tuberculosis and management of thimble bladder in 30-year male. (LAQ)**

- Kidney 80%; bladder disease descends from the kidney; thimble bladder <20 mL.
- Urine AFB culture: 3–5 early-morning samples; GeneXpert detects rifampicin resistance.
- 2HRZE/4HR; operate after 4–6 weeks of ATT.
- Augment if capacity <100 mL; substitute if <20 mL; needs creatinine <2 mg/dL and a good urethra.

**II-2. Describe anatomy of lymphatic drainage of testis and scrotum. Describe management of stage one seminoma of testis. (LAQ)**

- Right testis to interaortocaval; left to para-aortic; crossover only right to left.
- Scrotum to ipsilateral superficial inguinal nodes; never operate trans-scrotally.
- Risk factors: >4 cm and rete testis invasion (not LVI).
- Surveillance preferred; carboplatin AUC7 ×1 (TE19) or RT 20 Gy (TE18).

**II-3a. Oligospermia. (SAQ)**

- WHO 2021: 16 million/mL; severe <5 million/mL.
- Karyotype and Y-microdeletion if <5 million/mL.
- Repair palpable varicoceles only; never give testosterone.

**II-3b. DMSA Renal Scan. (SAQ)**

- DMSA binds proximal tubules: a static cortical scan at 2–4 h.
- Gold standard for acute pyelonephritis and scars.
- Scar only if the defect persists ≥6 months; normal DRF 45–55%.

**II-3c. Dutasteride. (SAQ)**

- Dual type 1 + 2 inhibitor; 0.5 mg; DHT down 90%; half-life about 5 weeks.
- Volume down 25%; AUR and surgery down about half; needs 6 months.
- Double the PSA; CombAT and REDUCE.

**II-3d. Penile Prosthesis. (SAQ)**

- 3-piece inflatable gives the best rigidity and flaccidity.
- Infection 1–3%; coated devices and no-touch technique; remove all parts.
- Satisfaction 80–90%; repairs 15% at 5 years.

**II-3e. CAPD (Continuous Ambulatory Peritoneal Dialysis). (SAQ)**

- Peritoneum is the membrane; glucose or icodextrin gives ultrafiltration.
- Tenckhoff catheter with two cuffs; Kt/V ≥1.7 per week.
- Peritonitis: cloudy fluid, WCC >100/µL, >50% neutrophils; give intraperitoneal antibiotics.

**II-3f. Uroflowmetry. (SAQ)**

- Voided volume ≥150 mL; Qmax is the key value.
- Plateau = stricture; intermittent = underactivity or straining.
- Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.